

Kuan-Hung Chen

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Education

University of Warwick

MSc in Statistics (*with Distinction*)

Focus: Machine Learning, Neural Networks, Statistical Inference, Model Evaluation

Coventry, UK

Sep 2024 — Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3. Focus: Econometrics, Game Theory

Taipei, Taiwan

Sep 2019 — Jun 2024

Research and Projects

MSc Dissertation, University of Warwick

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Built a vine copula-based probabilistic forecasting model in PyTorch that absorbs new data through quasi-Bayesian recursive updates, without retraining.
- On real-world weather data, cut forecast error (CRPS) by 24% vs the strongest baseline, training in under 3 minutes on a laptop CPU.
- Co-authored a contributed talk extending this work to spatio-temporal forecasting, presented at the *Environmental Modelling meets Infectious Diseases* workshop, University of Warwick, Apr 2026.

Pollparse

ALBERT-based zh-TW poll parser powering Sootow (*open source*)

Sep 2026

- Fine-tuned an ALBERT encoder with PyTorch and Transformers to parse free-form zh-TW poll text into titles, options, settings, and time, reaching 87% exact match on real user inputs.
- Exported to ONNX and deployed as the backend API of Sootow, a live polling app, running a 16 MB model at about 1 ms per request on a single CPU core.
- Archived the app's traffic and usage metrics in PostgreSQL, with SQL views to tune review thresholds from real data.

zh-TW NLP Product Categorizer

TF-IDF + SVC/MNB classifier served with FastAPI and Docker (*open source*)

Aug 2026

- Combined character- and word-level TF-IDF features to handle zh-TW product names that mix Chinese, English brand names, and model numbers.
- Classified noisy real-world listings into 84 categories at 70% top-1 and 85% top-3 accuracy.

Open Source

Timeful — 1.7k-star Go/Vue scheduling platform

Event ID hardening and rate limiting, merged upstream

Sep 2026

- Replaced predictable, time-seeded `math/rand` event IDs with `crypto/rand` and a wider alphabet, growing the keyspace from 3.2M to 24.8B so private event links can no longer be guessed (#274).
- Added per-IP rate limiting to the unauthenticated lookup endpoints, charging only failed lookups so legitimate traffic is never throttled (#275).

Experience

Market Research Intern

KANTAR Worldpanel

Taipei, Taiwan

Jun 2023 — Apr 2024

- Engineered 10+ Python and VBA data pipelines automating data inspection, transformation, and report generation over 10 datasets of consumer panel data, cutting report production time by over 20x and adopted across the analyst team.
- Analysed consumer purchase data for 10+ FMCG clients, turning market trend and demand forecast analyses into client reports and presentations, and owned the analytics for one major client account end to end.

Skills

Programming: Python, Go, C++, SQL, JavaScript, Ruby, R

Web Development: HTML, CSS, SvelteKit, React, Ruby on Rails

Machine Learning: PyTorch, Hugging Face Transformers, ONNX, scikit-learn, NLP, Model Evaluation, Time Series

Engineering: Docker, FastAPI, Git, CI/CD, Bash, Unit & Integration Testing